Installing an operating system

VirtualBox laboratory report
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Objective and scope

Configure and verify an operating system in VirtualBox, including virtual hardware, drivers, updates, security and performance settings. The existing prac2 Ubuntu installation was reused at the owner's request. No reinstallation, repartitioning or formatting was performed during this session.

Environment and final VM configuration

Item	Verified configuration
Host	ASUS TUF Gaming A15 FA506NC; Windows 11 Pro; 15.3 GiB RAM; 12 logical CPUs
Virtualization	Oracle VirtualBox 7.2.8; hardware virtualization enabled in firmware
Guest	prac2; Ubuntu 26.04.1 LTS, 64-bit
CPU and RAM	2 virtual CPUs; 4096 MB assigned RAM
Firmware and boot	EFI; installed system disk first; optical drive empty
Storage	25 GiB dynamically allocated VDI; SATA AHCI; GPT
Display	VMSVGA; 128 MB video memory; 3D acceleration disabled
Network	NAT; Intel PRO/1000 MT Desktop; DHCP address 10.0.2.15
Verified kernel	Linux 7.0.0-42-generic; 7.0.0-30 retained for recovery

Existing disk layout

Partition	File system	Size and mount point
/dev/sda1	FAT32 / vfat	About 1.05 GiB; /boot/efi
/dev/sda2	ext4	About 23.95 GiB; /

The existing layout was preserved. There is no separate /home partition and no configured swap. The root file system had about 16 GiB free before the updates. Installation and VM creation times were not measured because the existing VM was reused.

Procedure and observations

Changes completed

Recovery points. Created Before OS laboratory 2026-11-01 before changing the VM settings. It preserves the original disk and configuration: 6100 MB RAM, 6 CPUs and 16 MB video memory. Created OS laboratory completed 2026-11-01 after verification to preserve the updated system and settings.

Boot troubleshooting. Kernel 7.0.0-38 stalled in the initial RAM disk while waiting for root and EFI devices. Their UUIDs matched the disk metadata and /etc/fstab. Selecting the existing 7.0.0-30 kernel in GRUB reached the desktop. Kernel 7.0.0-42 was then installed from Ubuntu repositories and verified after a cold start.

Drivers and integration. Verified ahci storage, e1000 network, vmwgfx display, snd_intel8x0 audio and vboxguest kernel drivers. Installed virtualbox-guest-utils and virtualbox-guest-x11 7.2.6 from Ubuntu repositories. Verified the guest service after restart. Audio hardware and its driver were detected; audible playback was not tested.

Updates and security. Refreshed APT indexes, allowed automatic updates to finish, applied remaining software updates and installed the newer kernel. Enabled UFW at startup with deny-incoming and allow-outgoing defaults. Confirmed automatic updates, Asia/Qyzylorda time and active network time synchronization.

Performance and power. Retained the balanced power profile and disabled automatic AC suspend inside the guest. No user-defined startup applications were found; no system startup components were removed. CPU, memory, storage and network status were checked after the update.

Before and after

Item	Initial state	Final state
RAM / CPUs	6100 MB / 6 CPUs	4096 MB / 2 CPUs
Video memory	16 MB	128 MB
Boot	7.0.0-38 stalled	7.0.0-42 verified
Guest utilities	Not installed	Installed; service active
Firewall / zone	Inactive / UTC	Active / Asia/Qyzylorda

Final checks found zero failed services, no pending APT upgrades and no broken package dependencies. RAM usage was about 1.2 GiB with 2.1 GiB available; the root file system was 32% used with about 16 GiB free. Three one-second vmstat intervals showed 98-99% CPU idle.

No controlled timing benchmark was run with the original CPU and RAM allocation, so no numerical speed improvement is claimed. Command results are saved in /home/vboxuser/os-lab. The existing user files and host operating system were preserved.

Review questions

Questions 1 to 5

1 Why is configuring the correct boot order in a virtual machine important?

The boot order tells the firmware which device to start. The ISO is used for the initial installation, while normal starts should use the installed virtual disk. A wrong order may reopen the installer or cause a no-bootable-medium error.

2 What are the advantages of creating separate partitions during OS installation?

Separate partitions can isolate system files, user data and boot files, simplify maintenance or reinstallation, and support different file systems or quotas. They are not a backup. This VM has separate EFI and root partitions; the existing layout was preserved.

3 Why should chipset and system drivers be installed before other device-specific drivers?

They provide basic support for buses, controllers and related devices, helping the operating system identify dependent hardware correctly. In this Linux VM, the kernel already supplies the virtual chipset and controller drivers; guest integration packages add VirtualBox features.

4 How can you verify if all virtual hardware components are functioning properly?

Check that expected devices appear and have drivers, then test their functions. Linux tools include lspci -k, lsmod, lsblk, ip and system logs. Test network access, storage, display behavior and audio playback. On Windows, check Device Manager for warning symbols. Driver detection alone does not prove every function works.

5 What security measures should be implemented after OS installation?

Apply security updates, enable automatic updates, use a firewall, maintain appropriate accounts and permissions, and install software from trusted sources. Keep independent backups and create snapshots before major changes. For prac2, UFW was enabled and automatic updates were confirmed; the existing account was retained.

Application to this laboratory

The Linux path of the assignment was used. Windows installation and driver procedures were not applicable. Linux hardware support was provided primarily by kernel drivers. The existing EFI and ext4 partitions were sufficient for this installation.

Review questions continued

Questions 6 to 10

6 How does the installation process differ between Windows and Linux?

Both start from installation media, select language and storage settings, copy files and configure an account. Windows commonly uses NTFS and may request a product key. Ubuntu commonly uses ext4, offers Linux-specific partition and swap choices, and obtains most drivers from its kernel and repositories. Exact steps depend on the release.

7 What is the purpose of using an ISO image in VirtualBox?

An ISO represents installation media as a file. VirtualBox attaches it to a virtual optical drive so the guest can boot an installer or live environment without a physical DVD or USB drive. After installation, normal starts use the virtual hard disk.

8 What is the difference between a physical computer and a virtual machine?

A physical computer runs directly on its hardware. A VM runs through a hypervisor and sees virtual CPUs, memory, disks and devices backed by host resources. Several VMs can share a host, but their performance and capacity depend on the resources available on that host.

9 What are VirtualBox Guest Additions and what functions do they provide?

They are guest drivers and services that improve integration with VirtualBox. Depending on the guest and configuration, functions include display resizing, mouse integration, time synchronization, shared folders and clipboard sharing. Installation does not by itself prove that every integration feature is enabled or working.

10 Why is creating a virtual machine snapshot useful before performing major system changes?

A snapshot preserves a VM state that can be restored if an update or configuration change fails. It supports experiments and troubleshooting. Restoring it also discards later changes in the affected VM state, and a snapshot does not replace an independent backup.

References

Supplied laboratory brief: INSTALLING AN OPERATING SYSTEM. Practical results were recorded from prac2 during this session.

Oracle VirtualBox 7.2 Guest Additions

Ubuntu package management

Ubuntu automatic updates

Ubuntu firewall documentation